
Bishoksan Kafle, PhD

Strategic Technology Leader | Research Scientist | Digital Transformation |
Automated Reasoning, Program Analysis and Formal Methods

 Canberra ACT | Australian Citizen | Baseline Security Clearance (Exp: Aug. 2037)
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Professional Summary

- ★ Accomplished research and technology leader with a PhD in Computer Science and over 10 years of experience delivering AI, software engineering, formal methods, program analysis, DevSecOps, and enterprise integration solutions across government, academia, industry, and the non-profit sector, translating advanced research into operational capabilities that strengthen organisational performance, cybersecurity, national security, and executive decision-making.
- ★ Experienced leading multidisciplinary teams, managing multimillion-dollar research programs, delivering enterprise technology solutions, influencing technical strategy, mentoring researchers and engineers, and building trusted relationships with senior executives, government agencies, academia, and industry.
- ★ Recognised for excellence with the Department of Home Affairs Secretary Award (2025) for best practice in service delivery, program/project management, and technical leadership. Recipient of Best Paper award at the PEPM Symposium (2015) for innovative research in program analysis.

Selected Strategic Projects

- * **CCF Modernisation project (funded by the Department of Home Affairs)** At the department I led the section's migration from SVN to Azure DevOps (ADO), designing and implementing DevSecOps CI/CD pipelines for IBM technologies (ACE, WebSphere Application Server and others). I championed the end-to-end approach with other peers, optimised legacy workflows, automated change-management approvals, and shifted effort from people to tools—delivering measurable time and effort savings, stronger security and auditability, and a reusable delivery pattern now being adopted by other sections.
- * **ProCode Project (funded by Spanish Ministry of Research) – Senior Researcher:** This project aims to develop foundational methods and technologies—based on formal analysis and verification—to improve the reliability, efficiency, and certification of critical software systems. Collaborating with project partners, I led the development of methods and tools to verify software resource usage and safety properties of critical software systems, advancing research in abstract interpretation and automated reasoning. Delivered peer-reviewed publications and prototypes adopted in academia and industry, contributing to more reliable, predictable, and compliant mission-critical systems.
- * **Analysing Computer Arithmetic to Improve Software Reliability Project (funded by the ARC) – Senior Researcher:** The project aims to Develop sound verification tools that fully account for the realities of computer integer arithmetic—rather than idealised mathematical integers to ensure more accurate and reliable software analysis. Collaborating with project partners, I contributed to the automatic understanding of arithmetic in computer programs, developing a bit-precise solver for machine arithmetic and proving efficient methods for reasoning about floating-point computations. Designed techniques to compute preconditions ensuring error-free execution. These results used techniques such as SMT solving, automated reasoning, and abstract interpretation and remain the state-of-the-art in the field.
- * **ENTRA project (funded by EU FP7) – Research Lead:** The project aims to enable energy-aware software development by using advanced program analysis to predict energy consumption early in the design phase, supporting the creation of greener IT systems. I collaborated with XMOS Ltd. to design microchips with 2% improved energy efficiency, leveraging computer architecture, advanced software analysis, and energy-efficient compilation and optimisation, resulting in millions in savings.

Education & Qualifications

PhD in Computer Science (*Danish Govt. scholarship*) - Roskilde University, Denmark (2012 - 2016)

- Thesis: Components for automatic Horn clause verification (Adv: Prof. John P. Gallagher)

European Master's in Computational Logic (*Erasmus Mundus Scholarship*) - Dresden University of Technology, Germany, Free University of Bozen-Bolzano, Italy, and New University of Lisbon, Portugal (2010 - 2012)

- Thesis: Worst-case execution time of programs (Adv: Prof. Pedro Barahona & Franck Cassez)

Bachelor's in Computer Science (*Nepalese Govt. scholarship*) - Central University of Las Villas, Cuba (2004-2009)

Leadership, Collaboration & Communication

- **Research leadership & excellence** | 25+ peer-reviewed publications (incl. 2 books) and 10+ advanced research tools in AI, program analysis, verification, and rule-based reasoning. International leader in program analysis and automated reasoning with roles on 15+ program committees, 2+ program chairs, steering committee member of a conference series and regular reviewer for 5+ top journals.
- **Strategic Project Delivery & Impact** | Led and successfully delivered 6+ large-scale international projects funded by NASA, the EU, and govt. of Australia, Spain, and Italy, showcasing expertise in managing complex, multi-stakeholder and high-impact projects. Skilled in translating technical research into strategic policy insights for senior executives.
- **Community Engagement & Contribution** | ARC assessor, honorary researcher at the University of Melbourne, and co-founder of Nepal Development Society — shaping national research funding and priorities, mentoring emerging researchers and supporting development through research, collaboration, and mentorship.
- **Sharing knowledge and Mentorship** | Delivered 40+ scientific talks, supervised 20+ graduate students driving knowledge transfer, fostering innovation and nurturing the next generation of STEM leaders.
- **Collaboration & Facilitation** | Chaired conferences, facilitated strategic planning, and led delivery of impactful projects and training programs driving research impact and enhanced learning outcomes.

Technologies and skills

Programming Languages	Years of experience	Self-Rating	Middleware Technologies	Years of experience	Self-Rating
Procedural (Java, C, C++, Python, Pascal)	12+	★★★★	IBM MQ, IIB, IBM ACE, ESQL, Broker Flows	4+	★★★★
Functional (Ocaml)	4+	★★★★			
Logical (Prolog)	10+	★★★★★			
Web Technologies			Constraint Solving Techniques		
JEE, JSF, JSP, Hibernate, JavaScript, CSS, HTML, DBs	7+	★★★★	SAT/SMT Solving, Constraint Solving and Optimisation	8+	★★★★★
Software Verification Technologies			Document Processing		
Abstract Interpretation, Model Checking, Semantic Program Transformation, Static Analysis, Automated Reasoning	11+	★★★★★	Microsoft Packages, Latex	15+	★★★★★
Conference Management			Database		
EasyChair, HotCRP	6+	★★★★★	SQL, MySQL, Oracle, IBM DB2, SQL Server	10+	★★★★
Source Control			CI/CD		
SVN, Git, Gitlab	8	★★★★★	ADO, Gitlab	4+	★★★★★
IDE			Data Analysis		
NetBeans, Eclipse, VisualStudio, Jupyter Notebook	7+	★★★★	Python, SPSS	5+	★★★★

Relevant Professional experience

Integration Developer

2022 - Present

Department of Home Affairs

ACT, Australia

Contributing to critical national programs and initiative through data science, automation, software systems, and cross-functional collaboration in a high-stakes environment.

- ★ **Cybersecurity leadership** | Led our section's integration into departmental cybersecurity hardening efforts, aligning with the **Government's Essential 8**, improving risk posture and compliance across sensitive systems.
- ★ **Strategic Impact** | Delivered automated solution that significantly reduced manual workloads, improved operational efficiency and enhanced data-driven decision-making—contributing directly to program effectiveness and supporting national security objectives.
- ★ **Intelligence & Data Management** | Oversee data management for Cargo and PNRGOV systems, ensuring quality, compliance, and security. Cargo supports \$26B+ in annual revenue; PNRGOV enables real-time risk assessment and national security through advance passenger data.
- ★ **Capability building** | Mentored junior staff, Served as a recruitment panel member for APS5 and APS6 positions in Home Affairs, contributing to strategic hiring decisions and talent acquisition.

Co-Founder and Executive Committee Member

2013 - Present

Nepal Development Society (NeDS)

Providing strategic and technical leadership to advance sustainable development, public health, and environmental initiatives in Nepal—applying research to real-world impact.

- ★ **Strategic Leadership** | Helped define the organisation's vision and research agenda, contributing to long-term planning and measurable impact in governance, policy, and community resilience.
- ★ **Systemic positive impact** | Empowered local communities and improved governance by fostering collaboration between government, civil society, and the private sector, informing more inclusive policies and change in Nepal.
- ★ **Program Design & Implementation** | Lead public health and community development initiatives—including diabetes and hypertension prevention, sustainable urban farming, and rural health awareness programs—applying research insights to design and guide evidence-based interventions

Research Fellow in Computer Science

2016 - 2022

The University of Melbourne, and IMDEA Software Institute

Australia & Spain

Led and contributed to impactful research efforts at the intersection of software reliability, automated reasoning and program analysis aligned with Defence science objectives.

- ★ **Tool Development & Innovation** | Developed advanced research tools and participated in tools competition, significantly advancing the fields of software reliability, program analysis and verification globally.
- ★ **Securing funds and strengthening quality** | Secured and led \$2M+ in competitive research funding, expanding research capabilities, and enhancing the quality and impact of academic research.
- ★ **Research Impact** | Co-authored 8+ peer-reviewed papers on program analysis, verification and decision procedures focused on safety- and mission-critical systems and presented at conferences.

Visiting Research Scientist

NASA Ames Research Center, Intelligent system Division

Mar – Jun 2015

California, USA

- ★ **Learning and Development** | Gained experience in NASA avionics systems and conducted resource analysis research under renowned guidance, resulting (later) in two published articles and the development of a research tool.
- ★ **Collaboration** | Presented findings at NASA, fostering valuable collaborations with researchers. These contributions advanced the understanding of avionics software and strengthened connections within the research community.

Software Engineer

F1 Soft International

2009 - 2011

Nepal

Provided software development services and played a key role in developing [e-Sewa](#), a widely used payment gateway in Nepal, leveraging business insight and cutting-edge technologies.

- ★ **Systems Improvement** | Successfully enhanced over 5 applications by integrating SMS operations and developed more than 11 mobile applications, including a mobile version of e-Sewa.

Certifications & Training, Presentations and Publications

Certifications & Training:

- *Artificial Intelligence in Government: Fundamentals*, Australian Government (2024)
- *Building and Learning High Performing Teams*, APS Academy (2024)
- *Building Relationships and Engagement*, APS Academy (2024)
- *Effective Management Conversations*, APS Academy (2025)
- *Crafting Quality New Proposals – Practitioner*, APS Academy (2023)
- *Planning and Managing Change – APS Academy* (2024)
- *Workplace Communication in Practice*, APS Academy (2023)

Selected Presentations (of a total 40+):

- *Static Resource Analysis with Application to Avionics Systems* to 30+ research scientists and business leaders at NASA Ames Research Center, CA, US (2015)
- *Transformation-Enabled Precondition Inference* to 137+ research scholars for ICLP Conference, online (2021)
- *A Benders Decomposition Approach to Deciding Modular Linear Integer Arithmetic* to 149+ research scholars for SAT Conference, Melbourne Australia (2017)
- *RAHFT: A Tool for Verifying Horn Clauses Using Abstract Interpretation and Finite Tree Automata* to 190+ research scholars for CAV Conference, Toronto, Canada (2016)

Selected [Publications](#) (of total 25+):

- [Journal] Bishoksan Kafle, Graeme Gange, Peter J. Stuckey, Peter Schachte & Harald Sondergaard: *A Lightweight Approach to Nontermination Inference Using Constrained Horn Clauses*. Software and System Modeling (2024).
- [Book] Robert Glück & Bishoksan Kafle: *Logic-Based Program Synthesis and Transformation - 33rd International Symposium*, LOPSTR. LNCS 14330, Springer 2023, ISBN 978-3-031-45783-8.
- [Journal] Bishoksan Kafle, John P. Gallagher, Graeme Gange, Peter Schachte, Harald Sondergaard & Peter J. Stuckey: *An iterative approach to precondition inference using constrained Horn clauses*, TPLP (2018).
- [Journal] Bishoksan Kafle & John P. Gallagher: *Constraint Specialisation in Horn Clause Verification*, Science of Computer Programming (2017).